

B.TECH FIRST YEAR — ENGINEERING SCIENCES

Engineering Chemistry

Electrochemistry and Applications

UNIT II — Complete Study Module

Topics Covered:

Electrochemical Cells · Nernst Equation · Batteries · Fuel Cells
Corrosion Mechanisms · Corrosion Control · Electroplating

Prepared for: B.Tech 1st Year Students

Reference: Jain & Jain, Atkins' Physical Chemistry

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How to Use This Module

- Each section begins with a **Definition**, followed by detailed theory and mechanism.
- All key reactions are presented in highlighted **Formula Boxes**.
- SVG diagrams simulate exam-style labeled drawings — study them carefully.
- The Q&A section mirrors actual university exam patterns (2-mark and 5-mark).
- The Quick Revision Sheet provides last-minute formula recall.

INTRODUCTION TO ELECTROCHEMISTRY

1.1 Definition

Electrochemistry is the branch of chemistry that deals with the relationship between electrical energy and chemical reactions. It studies how chemical changes produce electricity (galvanic cells) and how electricity drives chemical changes (electrolytic cells).

1.2 Scope and Importance

Electrochemistry plays a vital role in modern technology, including energy storage (batteries), energy conversion (fuel cells), metal extraction, electroplating, corrosion control, and biosensors. Understanding electrochemical principles is essential for engineers in materials, chemical, and electrical disciplines.

1.3 Key Terminology

Term	Definition
Electrolyte	A substance that dissociates into ions in solution, enabling ionic conduction.
Electrode	A conductor (metal/non-metal) in contact with an electrolyte where redox reactions occur.
Oxidation	Loss of electrons at the anode: $M \rightarrow M^{n+} + ne^{-}$
Reduction	Gain of electrons at the cathode: $M^{n+} + ne^{-} \rightarrow M$
EMF (E_{cell})	Electromotive force — the potential difference driving the cell reaction (in Volts).
Standard Electrode Potential (E°)	Potential measured against standard hydrogen electrode (SHE) at 25°C, 1 M, 1 atm.
Faraday (F)	Charge of 1 mole of electrons = 96,485 C mol ⁻¹

1.4 Oxidation–Reduction (Redox) Reactions

All electrochemical processes involve simultaneous oxidation and reduction. The species losing electrons is oxidized (reducing agent) and the species gaining electrons is reduced (oxidizing agent).

GENERAL REDOX HALF-REACTIONS

Oxidation (Anode): $Zn \rightarrow Zn^{2+} + 2e^{-}$

Reduction (Cathode): $Cu^{2+} + 2e^{-} \rightarrow Cu$

Overall Cell: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}} = +0.34 - (-0.76) = +1.10 \text{ V}$$

1.5 Faraday's Laws of Electrolysis

First Law: The mass of substance deposited at an electrode is directly proportional to the quantity of electricity passed.

Second Law: For the same quantity of electricity, the masses deposited are proportional to their equivalent weights.

FARADAY'S LAW FORMULA

$$m = (M \times I \times t) / (n \times F)$$

Where: m = mass deposited (g), M = molar mass (g/mol)

I = current (A), t = time (s), n = number of electrons, F = 96485 C/mol

ELECTRODES AND ELECTROCHEMICAL CELLS

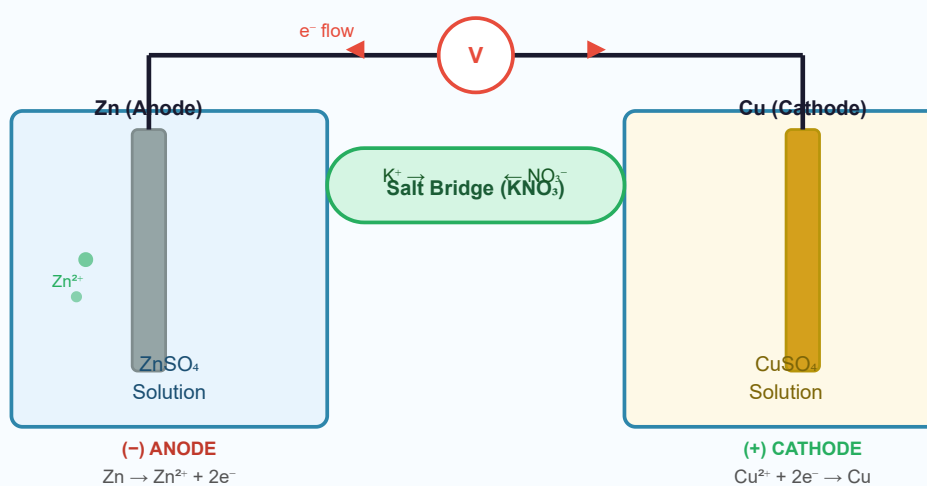
2.1 Types of Electrodes

Type	Description	Example
Metal-Metal Ion	Metal dipped in its ion solution	$\text{Zn} \mid \text{ZnSO}_4$
Gas Electrode	Gas bubbled over inert metal in ion solution	$\text{Pt} \mid \text{H}_2 \mid \text{H}^+$
Metal-Insoluble Salt	Metal coated with its insoluble salt	$\text{Ag} \mid \text{AgCl} \mid \text{Cl}^-$
Redox Electrode	Inert metal in solution of two oxidation states	$\text{Pt} \mid \text{Fe}^{3+}, \text{Fe}^{2+}$
Reference Electrode	Fixed, known potential; used as reference	SHE, Calomel electrode

2.2 Galvanic (Voltaic) Cell

A **galvanic cell** converts chemical energy into electrical energy spontaneously. It consists of two half-cells connected by a salt bridge. Oxidation occurs at the anode (negative terminal) and reduction at the cathode (positive terminal).

FIG 2.1 — DANIEL CELL (GALVANIC CELL) DIAGRAM



Electrons flow from Anode (Zn) through external circuit to Cathode (Cu). Ions migrate through salt bridge to maintain electrical neutrality.

2.3 Electrolytic Cell

In an **electrolytic cell**, electrical energy is supplied to drive a non-spontaneous chemical reaction. The positive terminal of the external battery is connected to the anode, and the negative terminal to the cathode.

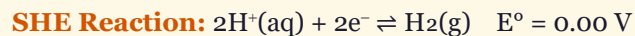
CELL NOTATION (DANIEL CELL)



$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = +0.34 - (-0.76) = +1.10 \text{ V}$$

2.4 Standard Hydrogen Electrode (SHE)

The SHE is the universal reference electrode with $E^\circ = 0.00 \text{ V}$. It consists of a platinum electrode immersed in 1 M HCl solution with H_2 gas bubbled at 1 atm pressure at 25°C .



NERNST EQUATION AND CELL POTENTIAL

3.1 Derivation of Nernst Equation

The **Nernst Equation** relates the electrode potential of a cell to the concentrations (activities) of the reacting species. It accounts for non-standard conditions of temperature and concentration.

For a general electrode reaction: $aA + ne^- \rightleftharpoons bB$

The Gibbs free energy change is: $\Delta G = \Delta G^\circ + RT \ln Q$

Since $\Delta G = -nFE$ and $\Delta G^\circ = -nFE^\circ$:

NERNST EQUATION — GENERAL FORM

$$E = E^\circ - (RT / nF) \times \ln Q$$

At 25°C (298 K), substituting $R = 8.314 \text{ J/mol}\cdot\text{K}$, $F = 96485 \text{ C/mol}$:

$$E = E^\circ - (0.0592 / n) \times \log Q \quad [\text{at } 25^\circ\text{C}]$$

Where:

E = Electrode potential under given conditions (V)

E° = Standard electrode potential (V)

R = Universal gas constant = $8.314 \text{ J K}^{-1} \text{ mol}^{-1}$

T = Temperature in Kelvin

n = Number of electrons transferred

F = Faraday's constant = 96485 C mol^{-1}

Q = Reaction quotient = $[\text{Products}] / [\text{Reactants}]$

3.2 Nernst Equation for Cell Potential

FOR THE DANIEL CELL: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - (0.0592 / 2) \times \log([\text{Zn}^{2+}] / [\text{Cu}^{2+}])$$

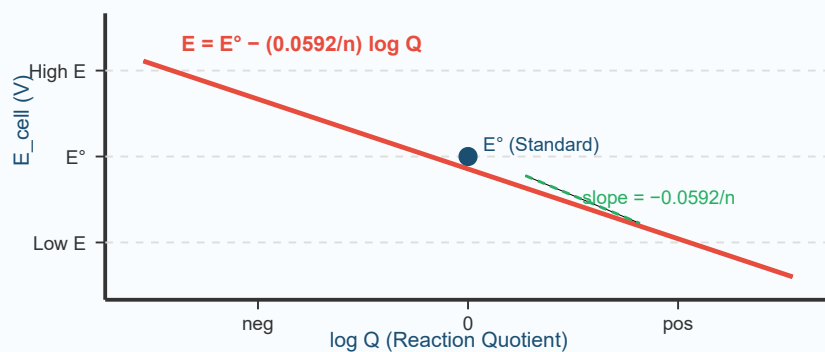
If $[\text{Zn}^{2+}] = 0.1 \text{ M}$, $[\text{Cu}^{2+}] = 1.0 \text{ M}$:

$$E_{\text{cell}} = 1.10 - (0.0296) \times \log(0.1/1.0)$$

$$E_{\text{cell}} = 1.10 - (0.0296) \times (-1)$$

$$E_{\text{cell}} = 1.10 + 0.030 = 1.130 \text{ V}$$

FIG 3.1 — CELL POTENTIAL VS LOG[Q] (NERNST EQUATION GRAPH)



As Q increases (more products), cell potential decreases linearly. At equilibrium $Q = K$, $E_{\text{cell}} = 0$.

3.3 Applications of Nernst Equation

- Calculation of electrode/cell potential at non-standard conditions
- Determination of equilibrium constant: $\log K = n \times E^\circ / 0.0592$
- pH measurement using glass electrode
- Calculating solubility products
- Understanding concentration cells (EMF generated by concentration difference)

NUMERICAL EXAMPLE 3.1

Problem: Calculate E_{cell} for: $\text{Zn} \mid \text{Zn}^{2+}(0.01 \text{ M}) \parallel \text{Cu}^{2+}(1.5 \text{ M}) \mid \text{Cu}$; $E^\circ_{\text{cell}} = 1.10 \text{ V}$, $n = 2$

Solution:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - (0.0592/n) \times \log([\text{Zn}^{2+}]/[\text{Cu}^{2+}])$$

$$E_{\text{cell}} = 1.10 - (0.0592/2) \times \log(0.01/1.5)$$

$$E_{\text{cell}} = 1.10 - 0.0296 \times \log(0.00667)$$

$$E_{\text{cell}} = 1.10 - 0.0296 \times (-2.176)$$

$$E_{\text{cell}} = 1.10 + 0.0644 = \mathbf{1.164 \text{ V}}$$

PRIMARY CELLS – ZINC-AIR BATTERY

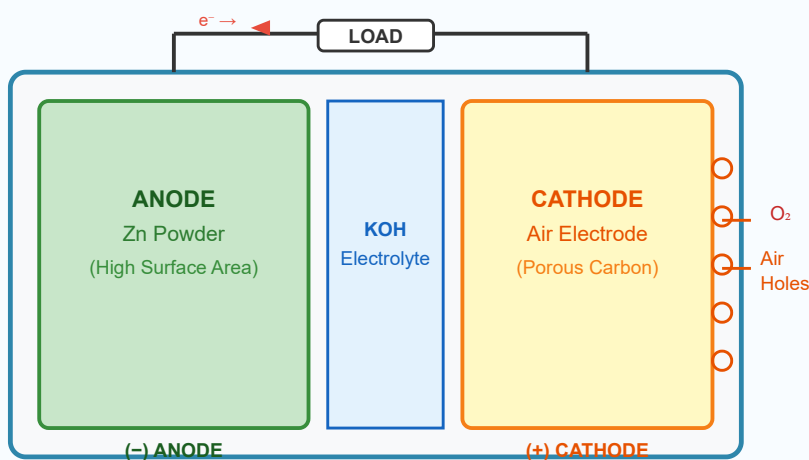
4.1 Definition

A **primary cell** is a battery that converts chemical energy to electrical energy through an irreversible electrochemical reaction. It cannot be recharged once discharged. The **Zinc-Air battery** uses zinc as the anode and atmospheric oxygen as the cathode reactant.

4.2 Construction and Working Principle

- **Anode:** Zinc powder (high surface area)
- **Cathode:** Porous carbon/air electrode (allows O₂ diffusion)
- **Electrolyte:** Concentrated KOH (potassium hydroxide) solution
- **Separator:** Ion-permeable membrane
- **Operating voltage:** 1.35–1.65 V per cell

FIG 4.1 – ZINC-AIR BATTERY STRUCTURE

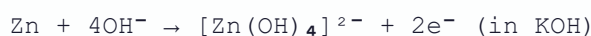


O₂ from air diffuses through porous cathode, reacts with KOH electrolyte, electrons flow through external circuit from Zn anode.

4.3 Electrochemical Reactions

ZINC-AIR BATTERY REACTIONS

Anode (Oxidation): $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$



Cathode (Reduction): $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$

Overall Reaction: $2\text{Zn} + \text{O}_2 \rightarrow 2\text{ZnO}$

Cell Voltage: ~1.4 V, Energy Density: 1,084 Wh/kg

4.4 Advantages and Limitations

Advantages

- Very high energy density (~1,084 Wh/kg)
- Lightweight — no heavy cathode material
- Flat discharge profile
- Environmentally friendly
- Long shelf life (sealed)

Limitations

- Not rechargeable (primary cell)
- Sensitive to humidity and CO₂
- Power output decreases in extreme temperatures
- Air access must be controlled

4.5 Applications

- Hearing aids (most common application)
- Medical devices and pagers
- Electric vehicles (Zn-air is being researched for EV use)
- Remote sensing equipment

SECONDARY CELLS – NICKEL-CADMIUM (NI-CD) BATTERY

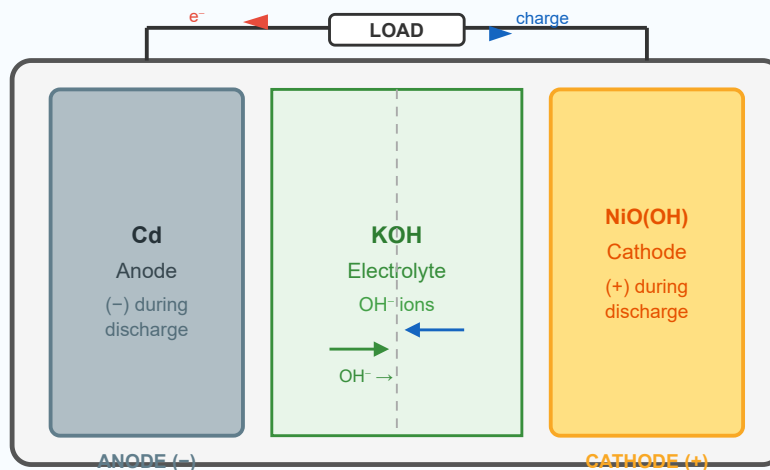
5.1 Definition

A **secondary cell** is a rechargeable battery in which the electrochemical reaction is reversible. The **Nickel-Cadmium (Ni-Cd)** battery uses nickel oxyhydroxide as cathode and cadmium as anode in KOH electrolyte.

5.2 Construction

- **Positive Electrode (Cathode during discharge):** Nickel oxyhydroxide — NiO(OH)
- **Negative Electrode (Anode during discharge):** Cadmium — Cd
- **Electrolyte:** 20–34% KOH solution
- **Separator:** Nylon or polypropylene membrane
- **Nominal voltage:** 1.2 V per cell

FIG 5.1 — NICKEL-CADMIUM BATTERY DIAGRAM



During discharge: Cd is oxidized at anode, NiO(OH) is reduced at cathode. During charging, reactions are reversed by external current.

5.3 Electrochemical Reactions

NI-CD BATTERY REACTIONS

DISCHARGE:



CHARGE (reverse) :



Nominal Voltage = 1.2 V

5.4 Advantages and Limitations

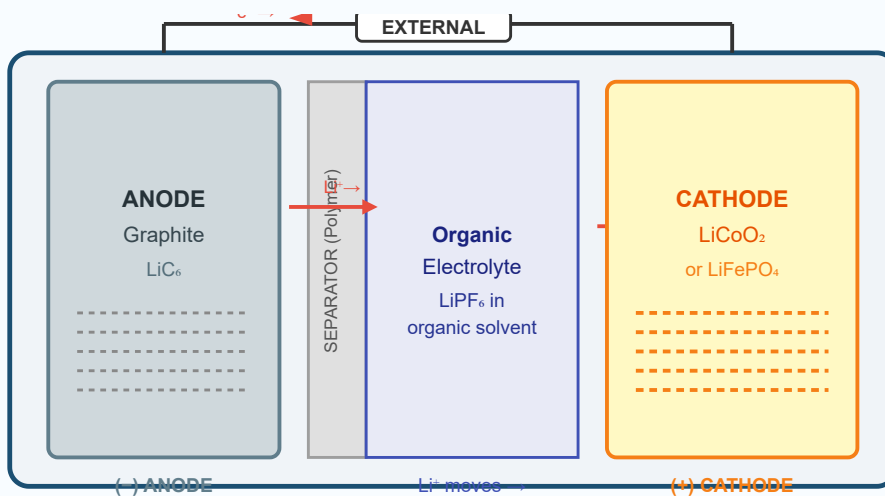
Property	Ni-Cd Battery
Rechargeability	500–1000 charge cycles
Voltage	1.2 V per cell
Energy Density	40–60 Wh/kg
Self-discharge	~20% per month
Memory Effect	Yes (major disadvantage)
Temperature Range	–20°C to +70°C
Toxicity	High — Cd is toxic, requires special disposal

LITHIUM-ION BATTERIES

6.1 Definition and Principle

A **Lithium-Ion (Li-ion) battery** is a rechargeable electrochemical energy storage device in which lithium ions (Li^+) move from the negative electrode (anode) to the positive electrode (cathode) during discharge through an organic electrolyte, and in the reverse direction during charging. The process is called **intercalation** — Li^+ ions insert/extract from host lattice structures without significant structural change.

FIG 6.1 — LITHIUM-ION BATTERY INTERNAL STRUCTURE

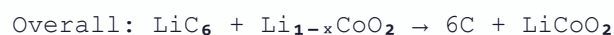
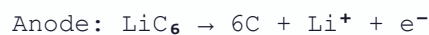


During discharge, Li^+ ions de-intercalate from graphite anode and intercalate into cathode (LiCoO_2). Electrons flow externally. During charging, process reverses.

6.2 Electrochemical Reactions

LI-ION BATTERY REACTIONS (LiCoO_2 | GRAPHITE)

DISCHARGE:



Cell Voltage: ~3.6–3.7 V

Energy Density: 150–250 Wh/kg

6.3 Comparison — Battery Types

Property	Zinc-Air	Ni-Cd	Li-Ion
Type	Primary	Secondary	Secondary
Voltage (V)	1.4	1.2	3.6–3.7
Energy Density (Wh/kg)	1,084	40–60	150–250
Rechargeability	No	500–1000 cycles	300–500 cycles
Memory Effect	N/A	Yes	No
Self-discharge/month	Very low	~20%	~2–5%
Toxicity	Moderate	High (Cd)	Low
Applications	Hearing aids	Power tools	Smartphones, EVs

FUEL CELLS – HYDROGEN-OXYGEN CELL

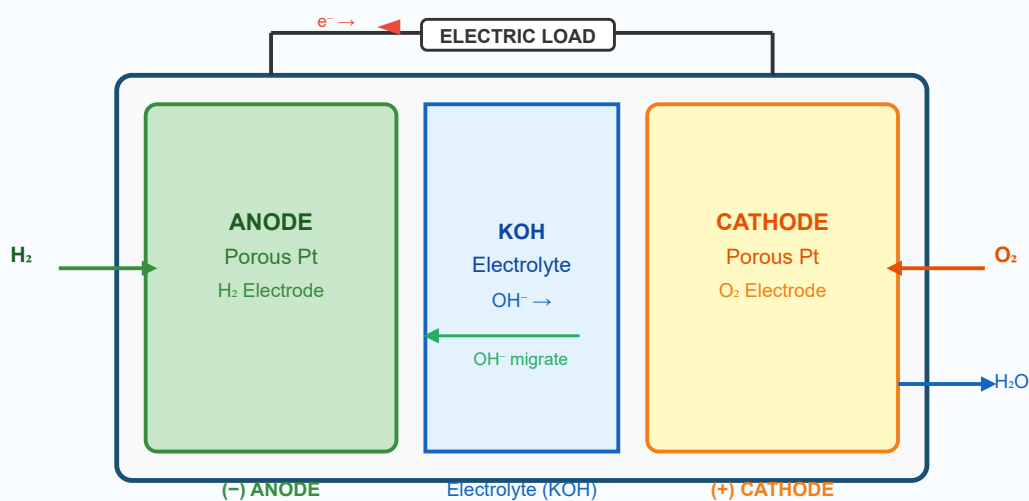
7.1 Definition

A **fuel cell** is an electrochemical device that continuously converts chemical energy of a fuel (hydrogen, methanol, etc.) directly into electrical energy without combustion, as long as fuel and oxidant are supplied. Unlike batteries, fuel cells do not store energy; they generate it continuously.

7.2 Hydrogen-Oxygen Fuel Cell — Construction and Working

- **Fuel (Anode side):** Hydrogen gas (H_2) is continuously supplied
- **Oxidant (Cathode side):** Oxygen gas (O_2) or air is supplied
- **Electrolyte:** KOH solution (alkaline type) or Proton Exchange Membrane (PEM)
- **Electrodes:** Porous platinum or nickel catalysts
- **Operating voltage:** ~ 0.7 – 1.0 V per cell; efficiency ~ 60 – 80%

FIG 7.1 — HYDROGEN-OXYGEN FUEL CELL



H_2 is oxidized at anode releasing electrons; O_2 is reduced at cathode. Only by-product is water. Efficiency: ~ 60 – 80% .

7.3 Electrochemical Reactions

H_2 - O_2 FUEL CELL REACTIONS (ALKALINE KOH ELECTROLYTE)



Overall: $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$ ($\Delta G = -237 \text{ kJ/mol}$)

Theoretical EMF = 1.23 V

7.4 Advantages and Applications

Advantages

- High efficiency (60–80%)
- Zero harmful emissions (only H_2O)
- Quiet operation — no moving parts
- Continuous power as long as fuel supplied
- Scalable design

Applications

- Space vehicles (NASA Apollo program)
- Hydrogen-powered automobiles
- Stationary power plants
- Submarines and buses
- Portable power systems

INTRODUCTION TO CORROSION

8.1 Definition

Corrosion is the gradual degradation or deterioration of a metal or its properties due to chemical or electrochemical reaction with the surrounding environment (atmosphere, water, soil, or chemicals). Corrosion is essentially the reverse of metallurgy — metals return to their more stable oxide/salt states.

Economic Impact: Corrosion costs approximately 3–4% of global GDP (~\$2.5 trillion/year). It causes structural failures, pipeline bursts, vehicle deterioration, and infrastructure damage.

8.2 Types of Corrosion

Type	Environment	Mechanism	Example
Wet / Electrochemical	Aqueous (water)	Electrochemical cell formation	Rusting of iron
Dry / Chemical	Dry gases, high temperature	Direct chemical attack	Scaling of steel at high T
Galvanic	Electrolyte with two metals	Electrochemical, dissimilar metals	Fe-Cu couple in seawater
Differential Aeration	Varying O ₂ concentration	Concentration cell formation	Crevice corrosion
Stress Corrosion	Corrosive + mechanical stress	Stress + chemistry	Brass in ammonia
Pitting Corrosion	Aqueous chloride solutions	Localized breakdown of passive film	Stainless steel in seawater

8.3 Factors Affecting Corrosion Rate

- **Nature of Metal:** Reactive metals (lower in EMF series) corrode faster; position in galvanic series matters
- **pH of Environment:** Acidic conditions accelerate corrosion; strongly alkaline can protect some metals (passivation)
- **Temperature:** Higher temperature generally increases corrosion rate
- **Humidity:** Relative humidity above 60% accelerates atmospheric corrosion significantly
- **Dissolved Salts / Conductivity:** Higher ionic strength of solution increases corrosion (seawater is highly corrosive)
- **Surface Area and Nature:** Rough, porous surfaces with defects corrode faster
- **Presence of Oxygen:** O₂ acts as cathodic depolarizer, accelerating corrosion

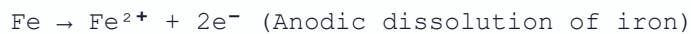
- **Mechanical Stress:** Stressed areas become anodic (stress corrosion cracking)
- **Microstructural Heterogeneity:** Grain boundaries, inclusions create local galvanic cells

ELECTROCHEMICAL THEORY OF CORROSION

9.1 Mechanism

The **electrochemical theory** proposes that corrosion of metals in the presence of an electrolyte (water with dissolved salts) occurs through the formation of a **local galvanic cell**. This requires: (a) an anode where oxidation occurs, (b) a cathode where reduction occurs, (c) an electrolyte connecting them, and (d) a metallic conductor between anode and cathode.

9.1.1 Anodic Reaction (Oxidation — Metal Dissolution)



9.1.2 Cathodic Reactions (Reduction — Depolarization)

IN ACIDIC CONDITIONS (HYDROGEN EVOLUTION):



IN NEUTRAL/ALKALINE CONDITIONS (OXYGEN ABSORPTION):



9.1.3 Secondary Reactions (Rust Formation)

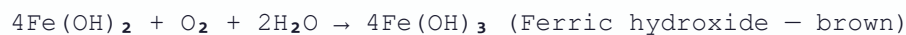
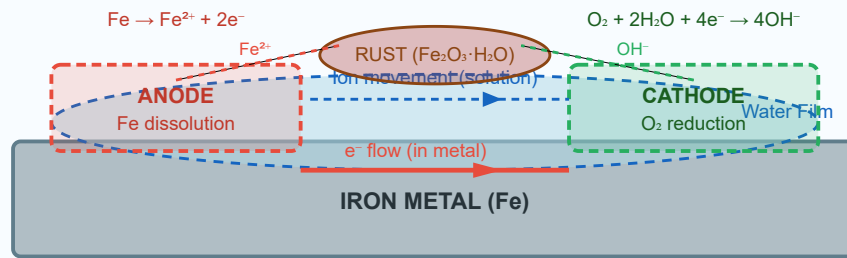


FIG 9.1 — ELECTROCHEMICAL CORROSION MECHANISM OF IRON

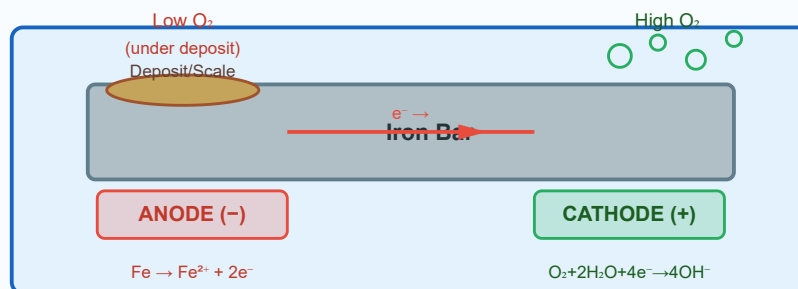


Local anode (Fe dissolution) and local cathode (O₂ reduction) form on same metal surface. Rust forms where Fe^{2+} meets OH^- ions.

9.2 Differential Aeration Corrosion

This occurs when different parts of the same metal are exposed to electrolyte with different oxygen concentrations. The area with **less oxygen becomes the anode** (corrodes), while the area with more oxygen becomes the cathode.

FIG 9.2 – DIFFERENTIAL AERATION CORROSION CELL



Area under deposit (low O₂) acts as anode and corrodes. Area exposed to air (high O₂) acts as cathode — no corrosion occurs there.

9.3 Galvanic Corrosion

Galvanic corrosion occurs when two dissimilar metals are electrically connected in an electrolyte. The less noble metal (lower in galvanic series / more active) acts as the anode and corrodes, while the more noble metal (cathode) is protected.

EXAMPLE: IRON–COPPER GALVANIC CELL IN SALTWATER

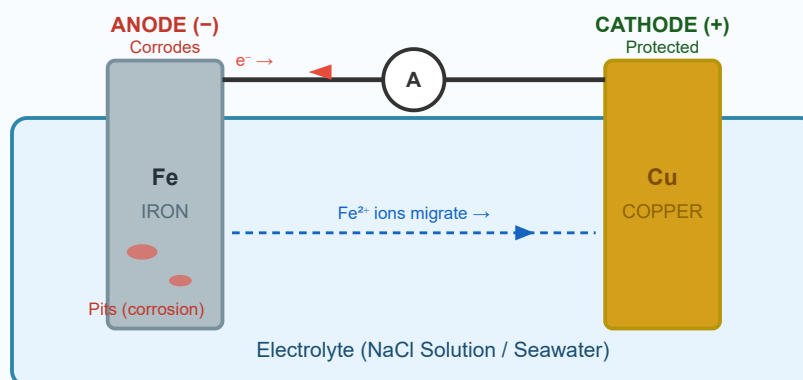
Anode (Fe – more active): $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$ [CORRODES]

Cathode (Cu – more noble): $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ [PROTECTED]

Rule: More active metal (lower SRP) = Anode = Corrodes faster

TYPES OF CORROSION – DETAILED ANALYSIS

FIG 10.1 – GALVANIC CORROSION SETUP (FE-CU COUPLE)



Iron (less noble, lower SRP = $-0.44V$) acts as anode and corrodes. Copper (more noble, SRP = $+0.34V$) acts as cathode and is protected.

10.2 Comparison of Corrosion Types

Feature	Galvanic Corrosion	Differential Aeration	Stress Corrosion
Cause	Dissimilar metals in electrolyte	Unequal O ₂ distribution	Stress + corrosive environment
Anode	Less noble (active) metal	Low O ₂ region	Stressed region
Cathode	More noble metal	High O ₂ region	Unstressed region
Example	Fe-Cu, Zn-Fe couples	Metal under water droplet center	Brass in ammonia
Prevention	Avoid dissimilar metals; insulate	Remove deposits; improve ventilation	Stress relief; annealing

10.3 Galvanic Series (Selected Metals)

Anodic End (Active — More likely to corrode): Mg → Zn → Al → Cd → Fe/Steel → Ni → Sn → Pb → H₂ → Cu → Ag → Pt → Au **:Cathodic End (Noble — Least likely to corrode)**

Metals further apart in the galvanic series exhibit more severe galvanic corrosion when coupled.

DRY CORROSION AND PILLING-BEDWORTH RATIO

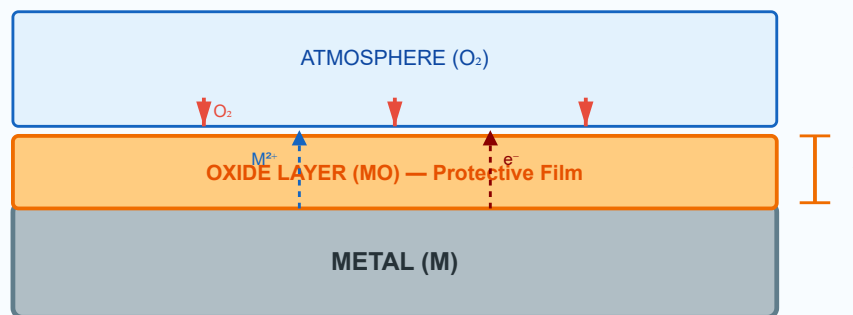
11.1 Dry Corrosion — Definition

Dry corrosion (chemical corrosion) occurs when metals react directly with gases (O_2 , H_2S , SO_2 , Cl_2) at high temperatures in the absence of moisture. The most common form is **oxidation**, where metals react with atmospheric oxygen to form a surface oxide film.

11.2 Oxide Film Formation Mechanism

- 1 Metal surface reacts with O_2 : $M + O_2 \rightarrow MO$ (oxide film forms immediately)
- 2 Initial film grows rapidly by electron and ion transport through the film
- 3 As film thickens, diffusion becomes rate-limiting — growth slows (parabolic law)
- 4 If film is protective, it acts as a barrier (Passivation). If non-protective, corrosion continues.

FIG 11.1 — OXIDE FILM FORMATION DURING DRY CORROSION



Metal ions (M^{2+}) diffuse outward through oxide layer while electrons travel through the film. O_2 reacts at oxide surface. Film grows with time.

11.3 Pilling-Bedworth (P-B) Ratio

The **Pilling-Bedworth Ratio** is the ratio of the volume of oxide formed to the volume of metal consumed during oxidation. It determines whether the oxide film formed is **protective or non-protective**.

PILLING-BEDWORTH RATIO FORMULA

P-B Ratio = Volume of Oxide / Volume of Metal consumed

P-B Ratio = $(M_{\text{oxide}} \times \rho_{\text{metal}}) / (n \times M_{\text{metal}} \times \rho_{\text{oxide}})$

Where:

M_{oxide} = Molar mass of oxide

M_{metal} = Atomic mass of metal

ρ_{oxide} = Density of oxide

ρ_{metal} = Density of metal

n = Number of metal atoms per formula unit of oxide

11.4 Interpretation of P-B Ratio

P-B Ratio	Oxide Film	Protection	Example
< 1	Porous, insufficient coverage	Non-protective (cracks)	Alkali metals (Na, K)
1 – 2	Compact, adherent, continuous	Protective (ideal)	Al_2O_3 , Cr_2O_3 , NiO
> 2	Under compressive stress, spalls off	Non-protective (peels)	WO_3 , MoO_3

NUMERICAL EXAMPLE 11.1 – P-B RATIO FOR ALUMINUM

Given: $\text{Al} \rightarrow \text{Al}_2\text{O}_3$; $M(\text{Al}_2\text{O}_3) = 102 \text{ g/mol}$; $M(\text{Al}) = 27 \text{ g/mol}$; $\rho(\text{Al}_2\text{O}_3) = 3.97 \text{ g/cm}^3$; $\rho(\text{Al}) = 2.70 \text{ g/cm}^3$;

$n = 2$

P-B Ratio = $(102 \times 2.70) / (2 \times 27 \times 3.97)$

= $275.4 / 214.38$

= **1.28** → P-B Ratio between 1 and 2 → **Protective oxide layer**

This explains why aluminum does not corrode significantly in air despite being reactive.

CORROSION CONTROL METHODS

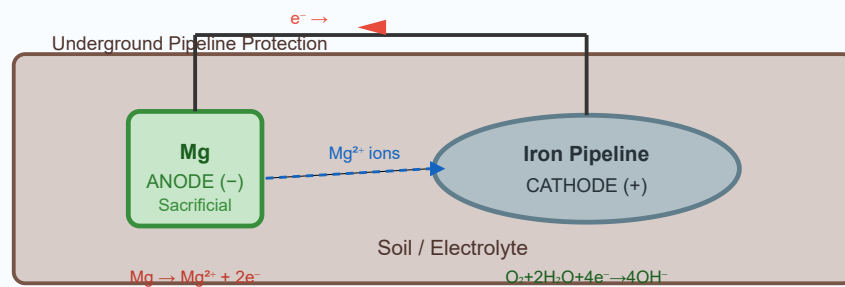
12.1 Cathodic Protection

Cathodic protection is a technique used to prevent corrosion by making the metal to be protected the **cathode** of an electrochemical cell, so that oxidation (corrosion) cannot occur at its surface.

Method 1: Sacrificial Anode Method

A more active metal (Zn, Mg, Al) is connected to the metal to be protected. The active metal sacrifices itself by acting as the anode and corroding, while the protected metal (Fe) acts as cathode.

FIG 12.1 — CATHODIC PROTECTION USING SACRIFICIAL ANODE



Mg sacrificial anode corrodes in place of iron pipeline. Electrons flow from Mg (anode) to Fe pipeline (cathode) through connecting wire.

Method 2: Impressed Current Method

An external DC current is applied to the metal so that it becomes the cathode. A rectifier converts AC to DC, and an inert anode (graphite or platinum) is used.

At protected metal (cathode): $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ [NO corrosion]

At external inert anode: $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

12.2 Anodic Protection

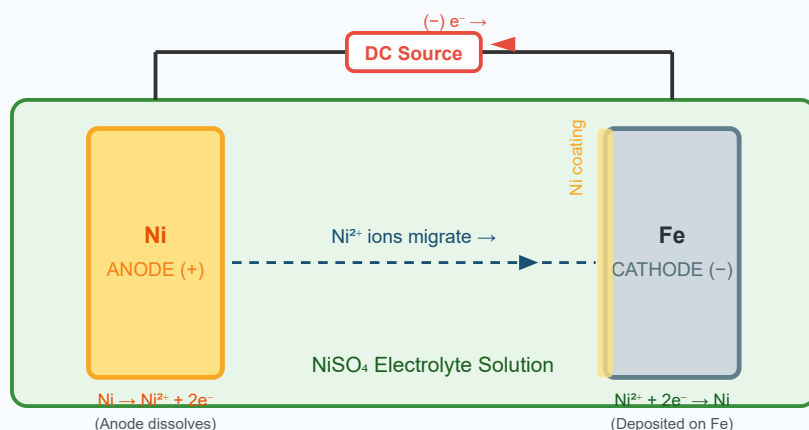
Anodic protection involves making the metal the anode by applying a positive potential, which causes formation of a stable, adherent **passive oxide film** on the metal surface. This passive film prevents further dissolution. It is suitable for metals that exhibit passivation behavior (e.g., iron, stainless steel, chromium in strong acids).

Key Principle: Applied anodic current maintains potential in the passive region of the metal's polarization curve. The passive oxide film (Cr₂O₃, Fe₂O₃) formed protects the underlying metal.

12.3 Electroplating

Electroplating is the process of depositing a thin layer of a desired metal on a base metal using electrolysis. The base metal acts as the cathode, the plating metal as the anode, and the electrolyte contains salt of the plating metal.

FIG 12.2 — ELECTROPLATING SETUP (NICKEL PLATING ON IRON)



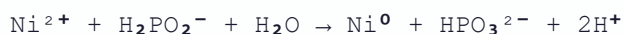
Ni anode dissolves continuously, maintaining Ni²⁺ ion concentration. Ni²⁺ deposits on Fe cathode, forming a bright corrosion-resistant nickel coating.

12.4 Electroless Plating

Electroless plating (autocatalytic plating) is a chemical deposition process that deposits a metal coating without using external electrical current. A reducing agent in the solution provides the electrons for metal ion reduction. The process is self-catalytic — the deposited metal itself catalyzes the reaction.

ELECTROLESS NICKEL PLATING REACTIONS

Using Hypophosphite (H₂PO₂⁻) as reducing agent:



(Ni is deposited; phosphorus may also co-deposit forming Ni-P alloy)

No external current needed — reducing agent provides electrons directly.

12.5 Comparison: Electroplating vs Electroless Plating

Feature	Electroplating	Electroless Plating
Power source	External DC current required	No external current (chemical)
Electron source	External power supply	Chemical reducing agent (e.g., H ₂ PO ₂ ⁻)

Substrate conductivity	Must be electrically conductive	Can coat non-conductors (plastics, ceramics)
Coating uniformity	Variable (edges/corners thicker)	Excellent (uniform on all surfaces)
Deposit quality	Pure metal	Alloy possible (Ni-P, Cu-P)
Cost	Moderate	Higher (expensive reducing agents)
Applications	Jewelry, automotive, electronics	PCBs, plastic parts, aerospace

12.6 Other Corrosion Control Methods

Method	Principle	Example
Alloying	Add corrosion-resistant elements to metal	Stainless steel (18% Cr, 8% Ni)
Painting/Coating	Physical barrier to prevent electrolyte contact	Bridge painting, automotive primers
Metal Cladding	Bonding a corrosion-resistant metal layer	Alclad (Al-clad duralumin)
Inhibitors	Chemicals that slow anodic or cathodic reaction	Chromates, nitrites, organic amines
Design Modification	Avoid crevices, water traps, dissimilar metals	Proper drainage design in structures
Hot-dip Galvanizing	Coat steel with Zn by dipping in molten Zn	Galvanized iron sheets, pipes

QUESTIONS AND ANSWERS

This section follows the university exam format. Answers are structured for maximum marks.

Part A — 2 Mark Questions

Q1. Define electrochemical cell and give an example.

2 Marks

An **electrochemical cell** is a device in which chemical energy is converted into electrical energy (galvanic/voltaic cell) or electrical energy is used to carry out a non-spontaneous chemical reaction (electrolytic cell). The conversion occurs through redox reactions at electrode-electrolyte interfaces.

Example: Daniel Cell — $\text{Zn} \mid \text{ZnSO}_4 \parallel \text{CuSO}_4 \mid \text{Cu}$ — where Zn acts as anode (oxidized) and Cu^{2+} is reduced at the cathode. EMF = 1.10 V. Oxidation occurs at anode ($\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$) and reduction at cathode ($\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$).

Q2. State and explain the Nernst Equation.

2 Marks

The **Nernst Equation** relates the cell potential to the standard electrode potential and the concentrations of reactants and products under non-standard conditions. It was derived by Walther Nernst in 1889.

The equation at 25°C is: $E = E^\circ - (0.0592/n) \times \log Q$, where E is actual potential, E° is standard potential, n is the number of electrons transferred, and Q is the reaction quotient = [products]/[reactants]. As Q increases (product concentration rises), the cell potential decreases. When $Q = K$ (equilibrium constant), $E_{\text{cell}} = 0 \text{ V}$ (cell is dead).

Q3. What is the Pilling-Bedworth ratio? What does it signify?

2 Marks

The **Pilling-Bedworth (P-B) Ratio** is the ratio of the volume of oxide formed to the volume of metal consumed during high-temperature oxidation (dry corrosion). It predicts whether the oxide layer formed will be protective or non-protective.

Formula: $P-B = (M_{\text{oxide}} \times \rho_{\text{metal}}) / (n \times M_{\text{metal}} \times \rho_{\text{oxide}})$. If $P-B < 1$: porous, non-protective film (e.g., Na). If $1 < P-B < 2$: compact, protective film (e.g., Al_2O_3 , $P-B = 1.28$). If $P-B > 2$: film under compression, spalls off, non-protective (e.g., WO_3). Thus it helps engineers select metals for high-temperature applications.

Q4. Distinguish between primary and secondary cells.

2 Marks

A **primary cell** is a non-rechargeable battery in which the electrochemical reaction is irreversible. Once the chemicals are consumed, the cell cannot be restored. Example: Zinc-Air battery, Leclanche cell. It has high energy density but cannot be recharged, making it suitable for hearing aids and remote controls.

A **secondary cell** is a rechargeable battery where the electrochemical reaction is reversible. By supplying electrical energy, the original reactants are regenerated. Examples: Ni-Cd battery, Lithium-ion battery, Lead-acid battery. Secondary cells have longer life cycles (hundreds to thousands of charge cycles) and are used in mobile phones, EVs, and power tools.

Q5. What is differential aeration corrosion?

2 Marks

Differential aeration corrosion occurs when different regions of the same metal are exposed to electrolyte solutions with different oxygen concentrations. The region with **lower oxygen concentration** acts as the anode and undergoes corrosion, while the region with higher oxygen concentration acts as the cathode and is protected.

Classic examples include: (1) a metal partially immersed in water — the submerged portion (low O_2) corrodes while the water-air interface (high O_2) acts as cathode; (2) metal under a water droplet — the center (low O_2) corrodes while the periphery (high O_2) does not. Removing deposits and improving aeration can prevent this type of corrosion.

5 MARK QUESTIONS

Q6. Explain the construction, working, and reactions of the Zinc-Air battery with a neat diagram. 5 Marks

Introduction: The Zinc-Air battery is a primary (non-rechargeable) electrochemical cell that uses zinc metal as the anode and atmospheric oxygen as the active cathode material. It has the highest energy density among all primary batteries (~1,084 Wh/kg), which makes it widely used in hearing aids and small medical devices.

Construction: (1) *Anode:* Zinc powder compressed into a gel with KOH electrolyte — high surface area maximizes reaction. (2) *Cathode:* Porous carbon-air electrode with a catalyst (MnO₂ or Pt) that allows oxygen from air to diffuse in. (3) *Electrolyte:* Concentrated KOH (potassium hydroxide) solution. (4) *Separator:* Ion-permeable polymer membrane prevents short-circuit. (5) *Air access holes* on the cathode side allow atmospheric O₂ entry.

Working Principle: When the battery is activated, zinc is oxidized at the anode, releasing electrons and Zn²⁺ ions. The electrons travel through the external circuit (powering the load) to the cathode, where atmospheric oxygen is reduced in the presence of KOH and water, producing hydroxide ions. These OH⁻ ions migrate through the electrolyte to complete the circuit.

Electrode Reactions:

Anode (-): $\text{Zn} + 4\text{OH}^- \rightarrow [\text{Zn}(\text{OH})_4]^{2-} + 2\text{e}^-$ (in KOH)

Cathode (+): $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$

Overall: $2\text{Zn} + \text{O}_2 \rightarrow 2\text{ZnO}$; Cell Voltage: ~1.35–1.65 V

Advantages: Highest energy density, lightweight (no heavy cathode material), flat discharge curve, environmentally friendly. **Limitations:** Not rechargeable, sensitive to CO₂ and humidity, air holes must remain open during use.

Q7. Derive the Nernst Equation and apply it to calculate cell potential. 5 Marks

Thermodynamic Basis: For any chemical reaction, the Gibbs free energy change is: $\Delta G = \Delta G^\circ + RT \ln Q$, where Q is the reaction quotient. For an electrochemical cell: $\Delta G = -nFE$ and $\Delta G^\circ = -nFE^\circ$, where n = moles of electrons, F = Faraday constant = 96,485 C/mol, E = cell potential.

Derivation: Substituting: $-nFE = -nFE^\circ + RT \ln Q$

Dividing by $-nF$: $E = E^\circ - (RT/nF) \ln Q$

Converting natural log to base-10 log: $E = E^\circ - (2.303RT/nF) \log Q$

At T = 298 K (25°C): $2.303 \times 8.314 \times 298 / 96485 = 0.0592$

Final Nernst Equation: $E = E^\circ - (0.0592/n) \log Q$

Numerical Application: For $\text{Zn} | \text{Zn}^{2+}(0.1 \text{ M}) || \text{Cu}^{2+}(0.01 \text{ M}) | \text{Cu}$, $E^\circ = 1.10 \text{ V}$, n = 2:

$$E = 1.10 - (0.0592/2) \times \log(0.1/0.01) = 1.10 - 0.0296 \times 1 = 1.10 - 0.030 = \mathbf{1.07\text{ V}}$$

Significance: The Nernst equation shows that cell potential depends on concentration — increasing product concentration decreases EMF. At equilibrium, $Q = K$ and $E = 0$. It enables calculation of equilibrium constants, solubility products, and pH via glass electrode.

Q8. Explain the electrochemical theory of corrosion with the mechanism of rusting of iron. 5 Marks

Introduction: The electrochemical theory of corrosion, proposed by Whitney (1903), states that corrosion of metals in the presence of electrolyte solutions occurs through the formation of local galvanic cells on the metal surface. The essential components are: anode (metal dissolving), cathode (where reduction occurs), electrolyte, and a metallic conductor between them.

Conditions for Rusting of Iron: Presence of moisture (electrolyte) and oxygen is essential. The iron surface is not perfectly homogeneous — areas of different stress, grain boundaries, impurities, and scale create potential differences, setting up local anodic and cathodic areas.

Mechanism Step by Step:

Step 1 — Anodic Reaction: At active areas (anodes), iron dissolves: $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$. These electrons flow through the iron metal to cathodic areas.

Step 2 — Cathodic Reaction (Oxygen Depolarization): At cathodic areas (well-aerated): $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$. The hydroxide ions are produced near cathodic areas.

Step 3 — Primary Product: Fe^{2+} (from anode) and OH^- (from cathode) migrate through the electrolyte and meet to form: $\text{Fe}^{2+} + 2\text{OH}^- \rightarrow \text{Fe}(\text{OH})_2$ (ferrous hydroxide — greenish-white, unstable)

Step 4 — Rust Formation: In the presence of more oxygen and water: $4\text{Fe}(\text{OH})_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{Fe}(\text{OH})_3 \rightarrow \text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$ (rust — red-brown, porous, non-protective)

Why Rust is Non-protective: Rust ($\text{Fe}_2\text{O}_3 \cdot \text{H}_2\text{O}$) is porous, voluminous (P-B ratio ~ 2.1), and non-adherent, so it does not shield the underlying iron from further attack. This leads to progressive corrosion until the entire metal is destroyed.

5 MARK QUESTIONS — CONTINUED

Q9. Explain the working of a Hydrogen-Oxygen Fuel Cell with reactions.

5 Marks

Introduction: A fuel cell is an electrochemical energy conversion device that converts the chemical energy of hydrogen fuel directly into electricity without combustion. Unlike batteries, fuel cells operate continuously as long as fuel (H_2) and oxidant (O_2) are supplied. They are more efficient (60–80%) than heat engines (30–40%) and produce no harmful emissions.

Construction: (1) *Anode*: Porous nickel or platinum electrode where H_2 is oxidized. (2) *Cathode*: Porous nickel or platinum electrode where O_2 is reduced. (3) *Electrolyte*: Hot concentrated KOH solution (alkaline cell) or proton exchange membrane (PEM cell). (4) *Catalyst*: Platinum dispersed on carbon maximizes electrode surface area for reactions.

Working: Hydrogen gas is continuously fed to the anode compartment. The Pt catalyst facilitates adsorption and ionization of H_2 . In alkaline electrolyte, H_2 reacts with OH^- ions from the electrolyte. Electrons released travel through the external circuit, generating electrical current. At the cathode, O_2 from air is reduced by electrons arriving from the circuit. In alkaline electrolyte, O_2 reacts with H_2O to regenerate OH^- ions. The overall reaction produces water as the only by-product.

Reactions (Alkaline KOH electrolyte):

Anode (–): $\text{H}_2 + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O} + 2\text{e}^-$

Cathode (+): $\frac{1}{2}\text{O}_2 + \text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{OH}^-$

Overall: $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$, $\Delta G = -237 \text{ kJ/mol}$, $E^\circ = 1.23 \text{ V}$

Advantages: Zero harmful emissions, high efficiency, quiet operation, no moving parts, continuous operation. **Applications:** NASA space vehicles, hydrogen cars (Toyota Mirai), stationary power plants, submarines, buses.

Q10. Explain Electroplating with the example of copper plating. Also explain electroless plating.

5 Marks

Electroplating — Definition: Electroplating is the process of depositing a thin, adherent layer of a desired metal (plating metal) on a base metal (substrate) using electrolysis, to improve appearance, corrosion resistance, hardness, or conductivity.

Basic Requirements: (1) DC power supply (2) Plating metal as anode (3) Object to be plated as cathode (4) Electrolyte = aqueous solution of salt of plating metal (5) Proper surface preparation — cleaning, degreasing, pickling

Copper Plating Example: Electrolyte = $\text{CuSO}_4 + \text{H}_2\text{SO}_4$ solution; Anode = pure copper plate; Cathode = object to be plated (e.g., iron, steel).

Anode reaction: $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$ (copper dissolves into solution)

Cathode reaction: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ (copper deposits on substrate)

The anode continuously replenishes Cu^{2+} ions consumed at cathode, keeping electrolyte concentration stable. Deposit quality depends on: current density (A/dm^2), pH, temperature, agitation, and presence of brighteners/additives.

Electroless Plating: Also called autocatalytic plating. No external current is used — a reducing agent in the bath provides electrons for metal deposition. The process is catalyzed by the deposited metal itself. It can coat non-conducting surfaces (plastics, ceramics). Key reaction: $\text{Ni}^{2+} + \text{H}_2\text{PO}_2^- + \text{H}_2\text{O} \rightarrow \text{Ni}^0 + \text{HPO}_3^{2-} + 2\text{H}^+$. Advantages: Uniform thickness on complex shapes, coats non-conductors, excellent corrosion resistance (Ni-P alloy). Used in printed circuit boards, hard disk drives, aerospace components.

Numerical Problems with Solutions

PROBLEM 1 — NERNST EQUATION

Q: Calculate the EMF of the cell: $\text{Cr} \mid \text{Cr}^{3+} (0.1 \text{ M}) \parallel \text{Fe}^{2+} (0.01 \text{ M}) \mid \text{Fe}$ at 25°C . Given: $E^\circ(\text{Cr}^{3+}/\text{Cr}) = -0.74 \text{ V}$; $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V}$

Solution:

Cell: Cr acts as anode (more negative E°), Fe as cathode

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = -0.44 - (-0.74) = +0.30 \text{ V}$$

Cell reaction: $\text{Cr} + \text{Fe}^{2+} \rightarrow \text{Cr}^{2+} + \text{Fe}$ (but $\text{Cr} \rightarrow \text{Cr}^{3+}$, so $3\text{Fe}^{2+} + 2\text{Cr} \rightarrow 2\text{Cr}^{3+} + 3\text{Fe}$, $n = 6$)

For simplification with given data ($n=2$ per Cr): $Q = [\text{Cr}^{3+}]^2 / [\text{Fe}^{2+}]^3 \rightarrow \text{complex}$. Using single electrode approach:

$$E_{\text{cell}} = 0.30 - (0.0592/6) \times \log([\text{Cr}^{3+}]^2 / [\text{Fe}^{2+}]^3)$$

$$= 0.30 - 0.00987 \times \log(0.01/0.000001) = 0.30 - 0.00987 \times 4 = 0.30 - 0.039 = \mathbf{0.261 \text{ V}}$$

PROBLEM 2 — FARADAY'S LAW

Q: How much copper (in grams) is deposited when a current of 2A is passed through CuSO_4 solution for 30 minutes? ($M = 63.5$, $n = 2$)

Solution:

$$t = 30 \times 60 = 1800 \text{ s}; \text{ Charge } Q = I \times t = 2 \times 1800 = 3600 \text{ C}$$

$$m = (M \times Q) / (n \times F) = (63.5 \times 3600) / (2 \times 96485)$$

$$m = 228600 / 192970 = \mathbf{1.185 \text{ g}} \text{ of copper deposited}$$

QUICK REVISION SHEET

Last-minute revision of all key formulas, reactions, and concepts from Unit II.

Key Formulas

Nernst Equation: $E = E^\circ - (0.0592/n) \log Q$ [at 25°C]

Standard Cell EMF: $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$

Gibbs Free Energy: $\Delta G = -nFE$ (spontaneous when $E > 0$)

Faraday's Law: $m = (M \times I \times t) / (n \times F)$

Equilibrium: $\log K = (n \times E^\circ_{\text{cell}}) / 0.0592$

P-B Ratio = $(M_{\text{oxide}} \times \rho_{\text{metal}}) / (n \times M_{\text{metal}} \times \rho_{\text{oxide}})$

Standard Electrode Potentials (Key Values)

- $\text{Li}^+/\text{Li} = -3.05 \text{ V}$ | $\text{Zn}^{2+}/\text{Zn} = -0.76 \text{ V}$ | $\text{Fe}^{2+}/\text{Fe} = -0.44 \text{ V}$
- $\text{Cd}^{2+}/\text{Cd} = -0.40 \text{ V}$ | $\text{Ni}^{2+}/\text{Ni} = -0.25 \text{ V}$ | $\text{H}^+/\text{H}_2 = 0.00 \text{ V}$
- $\text{Cu}^{2+}/\text{Cu} = +0.34 \text{ V}$ | $\text{Ag}^+/\text{Ag} = +0.80 \text{ V}$ | $\text{Au}^{3+}/\text{Au} = +1.50 \text{ V}$

Battery Summary

- **Zinc-Air (Primary):** Anode = Zn, Cathode = $\text{O}_2(\text{air})$, Electrolyte = KOH, $V = 1.4 \text{ V}$, ED = 1084 Wh/kg
- **Ni-Cd (Secondary):** Anode = Cd, Cathode = $\text{NiO}(\text{OH})$, Electrolyte = KOH, $V = 1.2 \text{ V}$, cycles = 500–1000
- **Li-Ion (Secondary):** Anode = LiC_6 , Cathode = LiCoO_2 , Electrolyte = LiPF_6 organic, $V = 3.6\text{--}3.7 \text{ V}$, ED = 150–250 Wh/kg

Fuel Cell Key Points

- Anode: $\text{H}_2 + 2\text{OH}^- \rightarrow 2\text{H}_2\text{O} + 2\text{e}^-$
- Cathode: $\frac{1}{2}\text{O}_2 + \text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{OH}^-$
- Overall: $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$; $E^\circ = 1.23 \text{ V}$; Efficiency = 60–80%
- No combustion, no emissions, continuous operation

Corrosion Key Points

- Rusting: $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$ (anode); $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$ (cathode)
- Rust: $4\text{Fe}(\text{OH})_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow \text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$
- Differential Aeration: Low O_2 = Anode (corrodes); High O_2 = Cathode
- Galvanic: Less noble metal = Anode = Corrodes
- P-B Ratio: <1 = porous; $1-2$ = protective; >2 = spalls off
- Al_2O_3 : P-B = 1.28 (protective); Rust (Fe_2O_3): P-B ≈ 2.1 (non-protective)

Corrosion Control Summary

- **Cathodic Protection:** Make metal cathode — sacrificial anode (Zn, Mg) or impressed current
- **Anodic Protection:** Apply anodic current → passive oxide film forms
- **Electroplating:** External DC current deposits metal from ion solution (Ni, Cu, Cr coatings)
- **Electroless Plating:** Chemical deposition, no current needed, uniform coating, coats non-conductors
- **Inhibitors:** Chromates (anodic), amines/organic compounds (cathodic/mixed)
- **Galvanizing:** Zn coating on steel — sacrificial protection

Important Comparisons at a Glance

Concept	Galvanic Cell	Electrolytic Cell
Energy conversion	Chemical → Electrical	Electrical → Chemical
Reaction type	Spontaneous ($\Delta G < 0$)	Non-spontaneous ($\Delta G > 0$)
Anode charge	Negative (−)	Positive (+)
Cathode charge	Positive (+)	Negative (−)
Example	Daniel cell, Batteries	Electrolysis, Electroplating

P-B Ratio	< 1	1–2	> 2
Film type	Porous / Cracked	Compact / Adherent	Spalls / Cracks
Protection	None	Excellent	Poor
Example	Na, K, Mg	Al, Cr, Ni	W, Mo, Fe

TEXTBOOKS AND REFERENCES

Primary Textbooks

1. Jain, P.C. and Jain, M. — *Engineering Chemistry*, 16th Edition. Dhanpat Rai Publishing Co., New Delhi, 2013. Chapters on Electrochemistry, Batteries, Fuel Cells, Corrosion and Corrosion Control are comprehensively covered. Recommended for B.Tech 1st year as per most Indian university syllabi.

2. Atkins, P., de Paula, J., and Keeler, J. — *Atkins' Physical Chemistry*, 10th Edition. Oxford University Press, 2010.

Chapter 7 (Chemical Equilibria) and Chapter 20 (Electrochemistry) provide rigorous derivations of the Nernst equation and electrode potential theory. Ideal for in-depth conceptual understanding.

Reference Books

3. Taylor, H.F.W. — *Cement Chemistry*, 2nd Edition. Thomas Telford Publishing, London, 1997.

Reference for corrosion in cementitious materials and construction chemistry.

4. Shaw, D.J. — *Introduction to Colloid and Surface Chemistry*, 4th Edition. Butterworth-Heinemann, Oxford, 1992.

Covers surface phenomena relevant to corrosion mechanisms and electrode-electrolyte interfaces.

5. Billmeyer, F.W. Jr. — *Textbook of Polymer Science*, 3rd Edition. John Wiley & Sons, 1984.

Reference for polymer-based coatings and membrane materials used in fuel cells and electroplating.

Additional Online Resources

- NPTEL Lecture Series — Engineering Chemistry (IITs and NITs) — nptel.ac.in
- Electrochemical Society (ECS) — electrochemsoc.org
- NACE International (Corrosion Engineering) — nace.org
- The Electrochemistry Encyclopedia — electrochem.cwru.edu

End of Unit II — Electrochemistry and Applications

All 14 sections covered · Diagrams included · Q&A with model answers · Quick revision sheet

Study systematically. Master the reactions. Draw the diagrams. Ace the exam!